

DravYantra Fleet Management Ecosystem

Executive Project Progress Report & Heavy Vehicle Pilot Action Plan

Generated: August 18, 2026 | Production Readiness Status: 72% Complete

EXECUTIVE SUMMARY

DravYantra is an end-to-end IoT platform for real-time fleet tracking, fuel theft monitoring, and driver scorecards. The software platform (Mobile App, AWS Amplify Admin Panel, Express Backend REST API, PostgreSQL DB) is 92% complete and production-tested. Hardware component selection is 100% finalized. The project is executing an Action Plan covering App fine-tuning & simulation testing (7 Days), hardware procurement & prototype development (20 Days), firmware debugging (2-3 Days), and commercial vehicle owner alignment & mechanical installation (20 Days).

PROJECT STATUS DASHBOARD

Overall System Readiness		72%
Software Apps & Simulation Scripts		92%
Hardware Component Selection (BOM)		100%
Hardware Procurement & Sourcing		0%
Firmware Prototype & Debugging		10%
Commercial Vehicle Mechanical Fitment		0%

FINALIZED HARDWARE BILL OF MATERIALS (BOM) & UNIT COSTS

Category	Component Model	Unit Price	Technical Function & System Role
Microcontroller	Nordic nRF52840-DK	Rs 5,000	Central ARM Cortex-M4F MCU, BLE 5.4, sensor data aggregator
GPS / GNSS	u-blox NEO-M9N-00B	Rs 1,500	Concurrent 4-GNSS constellation tracking, sub-meter positioning
Cellular Modem	u-blox SARA-R10	Rs 5,000	Multi-band LTE Cat 1 / NB-IoT cellular cloud transmitter
Vibration Sensor	Analog Devices ADXL345	Rs 250	3-axis accelerometer for Engine ON/OFF/IDLE & crash alert
Fuel Tank Sensor	Mercetech SP BLE-4	Rs 6,000	Wireless capacitive fuel level probe with BLE broadcast

TOTAL PROTOTYPE HARDWARE UNIT COST: Rs 17,750 per vehicle unit

1. DETAILED COMPLETED WORK (SOFTWARE, API & CLOUD PLATFORM)

STATUS	MODULE / SCOPE	DETAILED IMPLEMENTATION COMPLETED	COMPLETION
[X] Done	Mobile App	Flutter onboarding wizard, authentication, Firebase email link verification, protected router	100%
[X] Done	Mobile App	Real-time vehicle map tracking, trip history, fuel logs, vehicle & driver screens	100%
[X] Done	Admin Web	AWS Amplify deployment pipeline & multi-tenant Organization approval workflow	100%
[X] Done	Admin Web	Fleet Owners screen with default Status: All filter, soft delete & permanent purge	100%
[X] Done	Backend API	Node.js Express REST API, JWT auth, requireApprovedOrg gate & audit logging	100%
[X] Done	Database	PostgreSQL schema with audit logs, telemetry history & trip calculation engine	100%
[X] Done	Email & Reports	Transactional Nodemailer (X-Priority: 1) & automated ExcelJS/PDFKit report generators	100%
[X] Done	Hardware BOM	Finalized hardware component selection (Nordic nRF52840, NEO-M9N, SARA-R10, BLE-4)	100%

SOFTWARE ARCHITECTURE & SECURITY GATES (SYSTEM DESIGN IN EASY WORDS)

Organization Approval Security Gate When a new fleet owner signs up and verifies their email, our backend automatically blocks them from accessing app features until Super Admin approves their Organization Profile in the Admin Dashboard.	Permanent Database Cleanup When an admin permanently deletes a fleet owner, the backend runs a clean cascade across 10 database tables (vehicles, drivers, trips, fuel logs, audit logs) and deletes their account from Firebase Auth.
AWS Amplify Auto Web Hosting Our Admin Dashboard code is connected directly to GitHub. Whenever we push new updates to the repository, AWS Amplify automatically builds and publishes the new web version live on secure HTTPS servers.	PostgreSQL Trip & Telemetry Engine Our database continuously receives location updates from moving vehicles and automatically calculates total distance traveled, average speed, idling time, and fuel consumption.

2. FINALIZED HARDWARE COMPONENT STACK & SENSOR MECHANICS

Microcontroller MCU: Nordic Semiconductor nRF52840-DK 64MHz ARM Cortex-M4F MCU, BLE 5.4. Central hub reading wireless BLE fuel probe data and UART GNSS positioning strings.	Rs 5,000
GPS / GNSS Module: u-blox NEO-M9N-00B Concurrent satellite tracking across 4 GNSS constellations (GPS, GLONASS, Galileo, BeiDou) for sub-meter positioning accuracy.	Rs 1,500
Cellular Modem: u-blox SARA-R10 Multi-band LTE Cat 1 / NB-IoT cellular module with 2G fallback for low-latency HTTPS/MQTT data transfer to AWS cloud server.	Rs 5,000
Fuel Level Sensor Probe: Mercetech SP BLE-4 Wireless capacitive fuel probe installed in diesel tank to broadcast real-time fuel levels over BLE and detect sudden fuel theft.	Rs 6,000
Vibration & Motion Sensor: Analog Devices ADXL345 3-axis digital accelerometer measuring engine micro-vibrations across X, Y, Z axes for Engine ON/OFF/IDLE status & harsh braking.	Rs 250

3. ACTION PLAN TO HEAVY VEHICLE PILOT DEPLOYMENT

PHASE 1 (7 Days)	App Fine-Tuning & Backend Logic Simulation Testing Fine-tune minor UI details in Mobile App and Admin Dashboard. Run telemetry simulation scripts against backend API to thoroughly test trip calculation engines, speed alert thresholds, and fuel theft detection algorithms.
PHASE 2 (20 Days)	Hardware Component Procurement & Prototype Building Purchase physical components (Nordic nRF52840-DK, u-blox NEO-M9N, u-blox SARA-R10, ADXL345, Mercetech SP BLE-4). Study datasheets & SDKs, understand pinouts & protocols, and build initial working hardware prototype.
PHASE 3 (2-3 Days)	Hardware / Firmware Prototype Debugging Resolve hardware/firmware communication glitches, memory leaks, and BLE disconnection edge-cases. Calibrate ADXL345 3-axis vibration thresholds and map Mercetech BLE-4 fuel level capacitive lookup tables.
PHASE 4 (20 Days)	Commercial Truck Owner Alignment & Mechanical Fitment Coordinate with heavy commercial vehicle fleet owners and consult automotive mechanical engineers regarding physical IP67 enclosure mounting, 12V/24V power buck converter wiring, and fuel tank probe placement. Conduct live road trial (100+ km).